SMART INDIA HACKATHON 2026

LAYCAN

A freight decision engine for bulk cargo importers.

Freight stops being a daily purchase and becomes a managed position.

PS ID	26006
ORG	Ministry of Steel
DEPT	SAIL
THEME	Transport & Logistics

SAIL buys freight like a same-day plane ticket

Coal from Australia, Indonesia, Mozambique, the US and Russia into Paradip, Vizag, Gangavaram, Dhamra, Gopalpur, Haldia and Sandheads — one spot voyage at a time.

01 NO PRICE DISCIPLINE

Nobody can say whether today's offer is good relative to waiting. The decision defaults to “the plant needs coal — fix it.”

02 VESSEL CHOICE BY HABIT

The trade-off between economies of scale and a port's draft limit is rarely computed. Bigger looks cheaper until it can't berth.

03 NO INSTITUTIONAL MEMORY

Decisions live in email and spreadsheets. Nothing compares what was paid against what could have been paid.

04 FULLY EXPOSED TO PRICE RISK

Freight is carried unhedged, while the counterparty across every negotiation hedges routinely.

A hundred teams will build a forecasting model

It is the wrong product — and saying so out loud is where our credibility starts.

Dry bulk rates are close to a random walk

High volatility, volatility clustering, jumps, slow mean reversion. Beating a naive benchmark at 30–90 days is genuinely hard.

A liquid forward market already prices it

Forward Freight Agreements aggregate the informed opinion of every trading desk on earth. Six students will not out-forecast that curve.

So we don't try.

We benchmark against a random walk and against the curve, publish where we lose, and put the value somewhere it doesn't depend on forecast superiority.

The real problem is not prediction. It is decision-making under uncertainty.

Which is an optimal stopping problem, a constrained optimisation problem, and a risk management problem — wearing a forecasting costume.

Four questions a forecast can't answer

	A FORECASTING MODEL SAYS	LAYCAN SAYS
Fix today, or wait?	*“Rates may fall around 5%”*	Fix if it prints at or below $21.40/mt. It's at $23.10. Wait. Threshold rises to $22.60 by the 14th.
Which vessel class?	*“Capesize is cheaper per tonne”*	Kamsarmax. The Cape is draft-limited here, so you pay for capacity you cannot load.
Which instrument?	*— silent —*	60% of Q4 on a 4-voyage COA, 40% spot. Full cover looks cheaper but your volume band is ±18%.
How much risk?	*— silent —*	Unhedged Q4 exposure $14.2M, 95% CVaR $2.1M. 45 FFA lots cut tail risk 38% at zero premium.

Figures illustrative — replace with output from your own backtest and demo scenario.

One object: the Decision Memo

Every solver, every agent and every data source exists to produce this one page.

```
DECISION MEMO · SAIL-COK-2026-118
75,000 mt (±10% MOLOO) Coking Coal · Hay Point → Paradip
Laycan 05–15 Oct 2026 · 43 days to window open

RECOMMENDATION                          WAIT
Reservation rate today ....... $21.40 /mt
Market indication ............ $23.10 /mt  (+7.9%)
Threshold on 14 Sep .......... $22.60 /mt  (rising)
Latest advisable fixing date .. 21 Sep 2026

Vessel class ................. KAMSARMAX
Instrument ................... 4-voyage COA, 60% of Q4
Hedge ........................ sell 45 lots P5TC Q4

Expected landed ......... $22.05 /mt (P10 19.8 / P90 25.1)
vs fix-today ................ −$1.05 /mt = −$78,750
Confidence ................... MEDIUM-HIGH
Critic flags ................. 1 (cyclone watch)
```

WHAT MAKES IT A PRODUCT AND NOT A DASHBOARD

- → **One unambiguous action**
  not a chart to interpret
- → **A number that makes it falsifiable**
  you can check us tomorrow
- → **The cost of ignoring it**
  the counterfactual, in dollars
- → **A named objection**
  confidence with a reason attached
- → **The rejected alternative, with arithmetic**
  why not Capesize
- → **An expiry date**
  decisions go stale
- → **Full provenance on every figure**
  source, licence, timestamp

Illustrative scenario. The test: a procurement head can forward this to their director unedited.

The calculation the industry gets wrong

A port's draft limit is really a cargo limit. Ignore it and you recommend a vessel that pays for capacity it cannot load.

DRAFT → TONNES

```
d_max     = permissible_draft − UKC
ΔDWT      = TPC × 100 × (summer_draft − d_max)
DWT_avail = summer_DWT − ΔDWT
cargo     = DWT_avail − bunkers − water
            − stores − constants

FWA_mm = displacement / (4 × TPC)
DWA_mm = FWA_mm × (1025 − ρ_dock) / 25

INTAKE = min(weight_limited, volume_limited)
```

WORKED EXAMPLE · CAPESIZE AT A DRAFT-LIMITED PORT

Summer deadweight	180,000 mt
Summer draft	18.0 m
Permissible draft (illustrative)	14.5 m
Under-keel clearance	0.6 m
TPC	130 mt/cm
Deadweight sacrificed	−53,300 mt
Less constants and bunkers	−6,500 mt
MAX INTAKE	**≈ 120,200 mt**

67% of the ship you are paying for. Scale economy gone — and full intake means lightering at anchorage, which costs money and days.

Feasibility is a hard constraint, never a penalty. A vessel that cannot berth is not a worse answer — it is a wrong answer.

Draft figure illustrative pending verification (VERIFY-FIRST V3). Method is exact.

Two layers, one rule that makes it trustworthy

The reasoning layer decides what to ask. The deterministic core decides what is true.

REASONING LAYER · LLM agents on a typed graph

Gathers · interprets · challenges · explains · narrates

MAY NOT COMPUTE. MAY NOT EMIT A NUMERAL.

Enforced by a Pydantic validator and a CI check. Not by convention.

typed tool calls ↓ ↑ typed structured results

DETERMINISTIC CORE · pure Python, no LLM, unit- and property-tested

`physics/`	draft-limited intake, TPC, FWA, load lines	`voyage/`	TCE, bunkers (cube law), laytime, demurrage
`rates/`	OU-jump simulator, calibration, conformal bands	`timing/`	LSMC optimal stopping → reservation curve
`routing/`	marine graph, A*, weather-weighted edges	`assign/`	MILP over cargo × class × port × window
`risk/ hedge/`	CVaR, scenarios, hedge ratio, basis risk	`backtest/`	policy replay vs naive vs oracle

Every numeral in the product comes from a versioned solver and carries provenance to its source, licence and timestamp. A hallucinated figure is structurally impossible, not merely unlikely.

Agents at the edges, solvers in the middle

If you cannot explain what an agent does that a function call could not, delete the agent.

Market Analyst

rate state, curve, calibration

Fleet Supply

tonnage tightness in ballast radius

Demand

coal stocks, plant burn, volume band

Port Feasibility

hard physics, intake caps, reasons

Voyage Economist

TCE and full cost breakdown

Risk & Disruption

events → jump intensity, waiting days

Instrument Strategist

spot / TCT / COA / FFA mix

Hedge Desk

ratio, lots, effectiveness, basis risk

THE CRITIC

An agent whose only job is to attack the recommendation before it ships.

- Out-of-distribution state
- Regime change tests
- Forward-curve disagreement
- Single-source ablation
- Calibration drift
- Independent feasibility recheck
- Magnitude plausibility

When it can't get comfortable, the product says so and escalates to a human.

Fix-or-wait is an optimal stopping problem

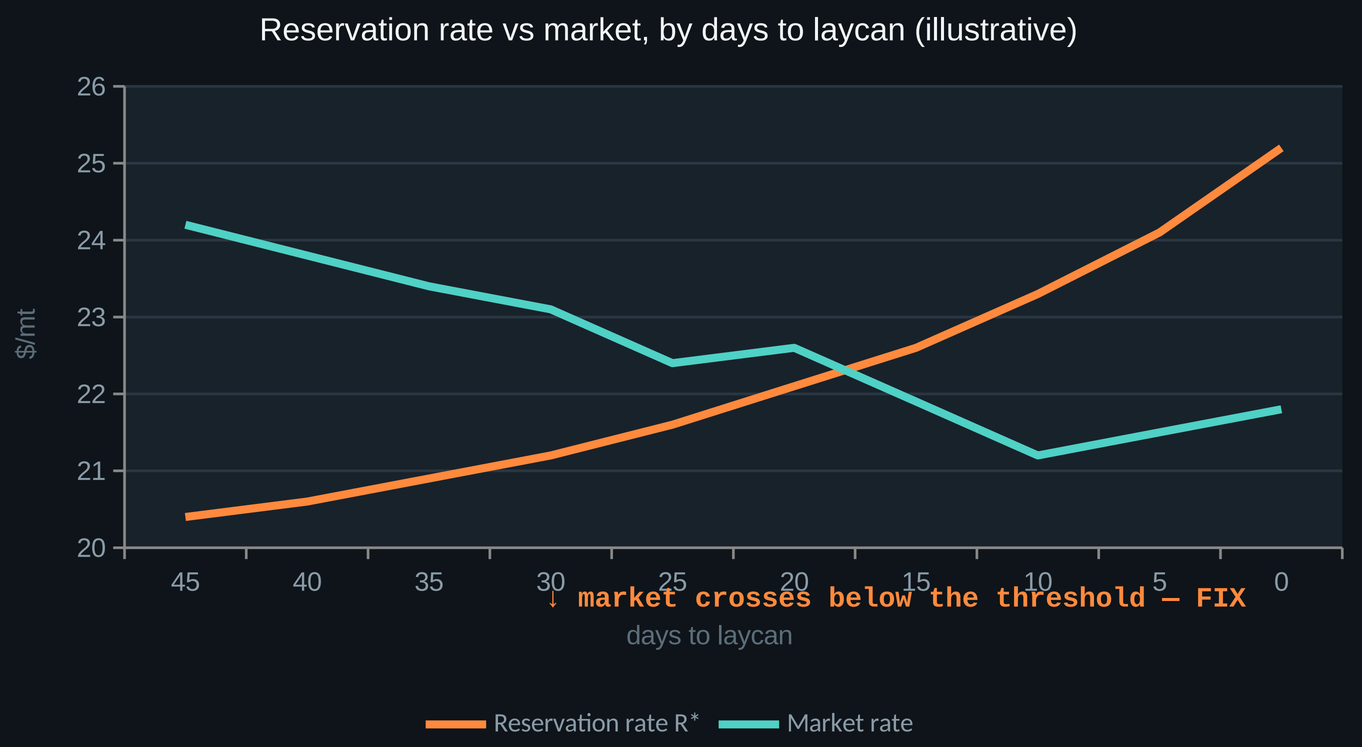
Structurally the same as exercising an American option — and solved the same way, with Least-Squares Monte Carlo.

```
V_t = min{ R_t , E[ V_t+1 | F_t ] }

R*_t = E[ V_t+1 | F_t ]

FIX  iff  R_t ≤ R*_t
```

The reservation rate is the continuation value. Two properties we test for, because they are how a real charterer behaves:

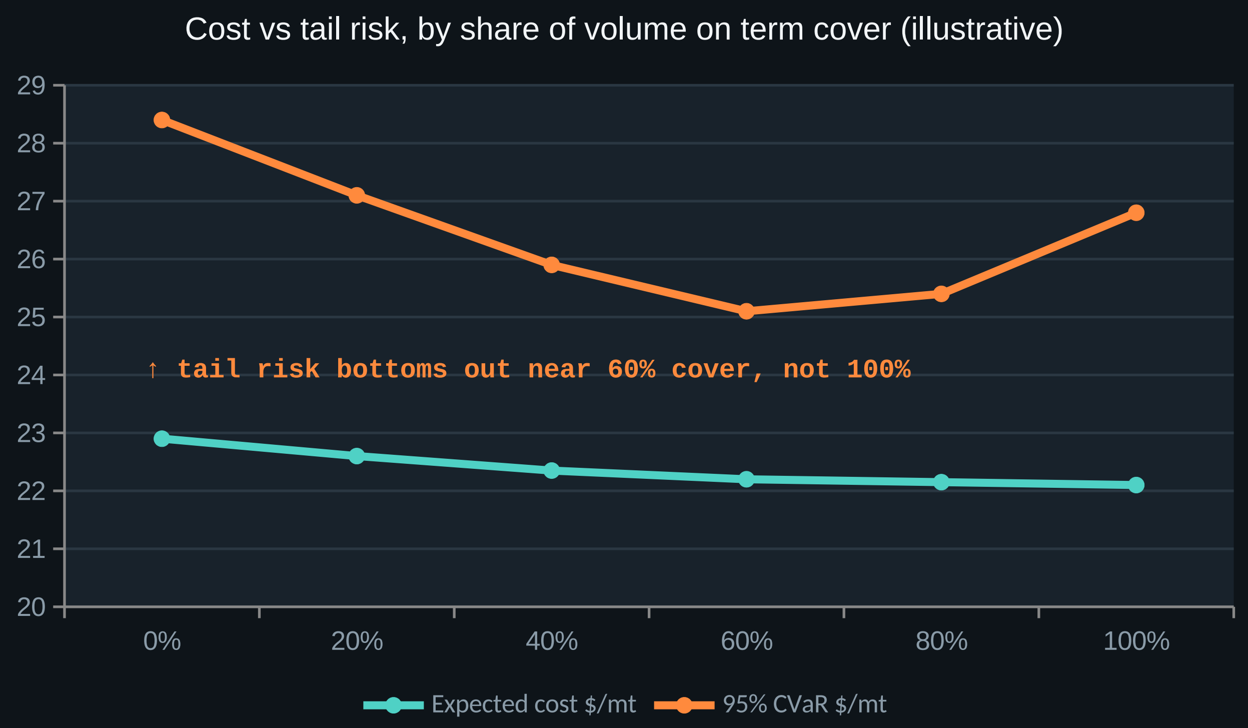
It rises as the laycan closes.
You get less choosy as you run out of runway.

It widens with volatility.
More uncertainty means more value in waiting.



The layer nobody sells to physical importers

Indian industrial buyers are structurally unhedged on freight while every counterparty they negotiate with hedges routinely.



WHY COVER IS CAPPED, NOT MAXIMISED

A term contract is only cheap if you can actually lift it. So COA coverage is capped by the lower confidence bound of the volume forecast — not the mean. Short-lifting costs more than the rate saving.

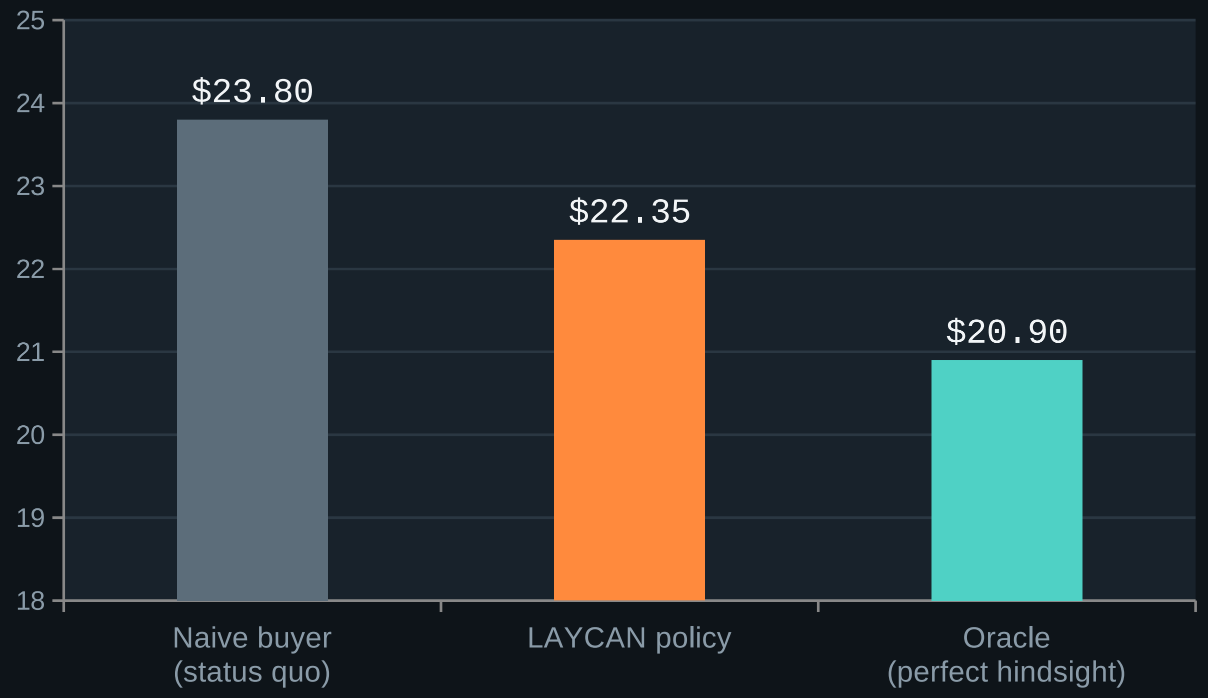
THE HEDGE, STATED HONESTLY

$h^* = \mathrm{Cov}(\Delta S, \Delta F)/\mathrm{Var}(\Delta F)$	minimum-variance ratio
$\text{effectiveness} = \rho^2$	variance actually removed
$\sigma(S - h^* F)$	residual basis risk, in $/mt

A hedge sold as perfect is one that will eventually embarrass you.

We backtest decisions, not forecasts

Everyone shows forecast error. We show money. Walk-forward, point-in-time features, a leakage test in CI, and net of all frictions.



CAPTURE RATIO

50% of the timing value that was theoretically available

Saving vs naive	$1.45 /mt
Regret vs oracle	$1.45 /mt
Worst quarter	always reported
Tail risk with hedge	CVaR −38%

Capture ratio is the number to lead with. It concedes the oracle is unreachable, and it is a claim a CFO can act on.

Illustrative structure — populate with your own backtest output. Never quote a saving without its worst case.

Our data honestly, before you ask

REAL AND CITED

- ✓ Daily port calls and trade estimates
- ✓ Official Indian port turnaround and pre-berthing detention
- ✓ Daily power-plant coal stocks
- ✓ Commodity and energy price series
- ✓ Marine weather and cyclone history
- ✓ Global news event feed
- ✓ Machine-readable port specifications
- ✓ Marine routing network graph
- ✓ Quarterly TCE by class from owners' filings

SIMULATED, AND BADGED AS SUCH

Licensed freight route assessments are paywalled and we will not scrape them. So the daily rate series is a calibrated stochastic process — and every simulated value carries a visible badge in the UI.

CALIBRATED TO

- Official quarterly TCE disclosures
- Freight-linked traded instrument volatility
- Published mean-reversion half-lives
- Event frequency from the news feed

Then validated against free observables, and the correlation table published.

WHAT WE REFUSE TO CLAIM

- ✗ That we beat the forward curve at long horizons
- ✗ A savings figure without its worst-case counterpart
- ✗ Simulated series presented as observed
- ✗ A confidence number whose calibration we haven't measured

A licence is a config change and a line in the funding plan — not an architectural problem.

Incumbents sell terminals. We sell a decision.

MARITIME INTELLIGENCE PLATFORMS

Vessel tracking, market data, fixture history, voyage management, chartering workflow

SOLD TO

Shipowners · operators · brokers · commodity trading desks

People who charter every day, have a desk, and already know what the market is worth. They need data, and they buy it by the seat.

They do not sell a prescriptive fix-or-wait threshold, a port-constrained vessel choice, an instrument mix, or a hedge — to a buyer with no desk.

LAYCAN

A daily decision with a threshold, a vessel, an instrument and a hedge — backtested in money

SOLD TO

Industrial importers moving 0.5–10 Mtpa with no chartering desk

Steel, power and cement companies with real freight exposure, no quantitative rate policy, and no hedge. Big enough for the savings to matter, small enough not to have built the capability.

Different buyer. Different product. An unoccupied position.

Be the decision to earn the data

Then be the data to earn the index — a benchmark for India-bound bulk freight that nobody publishes today.

PHASE 1 — 0–6 months

One route, one customer, shadow mode

Run alongside an existing desk for a quarter. Log every recommendation immutably. The deliverable is not software — it is a number.

PHASE 2 — 6–18 months

The product

Full instrument layer, hedge advisory, ERP integration, multi-plant portfolio. Three to five paying customers. Pricing normalises.

PHASE 3 — 18–36 months

The data asset

Models trained on real fixture outcomes. And an authoritative benchmark for India-bound dry bulk freight, which does not exist publicly today.

THE MOAT

Software features are copyable. Two things are not.

Every recommendation, stored with its model version, joined to what actually happened — the rate fixed, the days waited, the demurrage paid. Nobody else has this for India-bound bulk, because nobody else is in the decision loop.

And the benchmark. Baltic-grade assessments exist for the world's liquid routes, not for Newcastle → Paradip. A platform inside enough Indian importers' fixture flow can construct the reference nobody publishes — and index businesses become infrastructure.

What we built, and what we deliberately did not

Naming your exclusions is a credibility signal, not a weakness.

BUILT FOR SIH

- ✓ Canonical port, vessel and cargo domain model
- ✓ Rate state with basis decomposition
- ✓ Calibrated probabilistic forecasts
- ✓ Reservation-rate timing policy
- ✓ Draft-limited cargo intake solver
- ✓ MILP vessel and port assignment
- ✓ Full voyage economics and TCE
- ✓ Congestion and waiting nowcast
- ✓ Idle and repositioning comparison
- ✓ Instrument portfolio on a CVaR frontier
- ✓ Hedge sizing with basis risk
- ✓ Risk and disruption early warning
- ✓ Decision memo with full provenance
- ✓ Decision backtest harness

DELIBERATELY OUT OF SCOPE

Real-time global satellite AIS

commercial only — we use free coastal and port-call data and say so

Licensed route assessments

paywalled, and we will not scrape

Trade execution and clearing

we advise; a human and a broker execute

Charter party and laytime workflow

adjacent, valuable, a different product

Container, tanker, gas, breakbulk

dry bulk only — focus is the point

Rule: if it does not change the decision memo, it is not built before SIH.

Freight stops being a daily purchase and becomes a managed position.

A threshold, not an arrow

optimal stopping over the laycan window

Port physics as a constraint

draft-limited intake, not a lookup table

An instrument and a hedge

the layer nobody sells this buyer

Backtested in money

capture ratio, with the worst quarter shown